

Aura: Evaluating Tiny ML for Edge Deployment

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Abstract—The increasing computational and memory demands of state-of-the-art large language models (LLMs) pose significant barriers to their deployment in resource-constrained environments. This paper addresses the challenge of achieving high-quality, story-focused text generation using a significantly compressed LLM tailored for interactive narrative generation in gaming environments. By combining quantization and knowledge distillation techniques, we compress the Llama 3.2 1B parameter model to under 0.5 GB while preserving performance metrics such as perplexity, readability, and token diversity. Experimental results indicate that the optimized models achieve consistent readability and improved latency, with only minor trade-offs in coherence and diversity. This research contributes to the growing field of TinyLLMs by demonstrating practical methods to balance efficiency and storytelling quality, paving the way for real-time, on-device generative applications in memory-constrained platforms.

Index Terms—Quantization, Knowledge Distillation, Model Compression, TinyML

I. INTRODUCTION

The field of large language model (LLM) development has rapidly evolved in recent years. Early work focused on relatively simple tasks such as semantic classification and sentence completion[11]. Today’s LLMs achieve remarkable performance in a wide range of modalities and applications, including interactive textual dialogue[10], code generation, and multi-modal content creation[2]. However, these advancements have come with steep computational and storage demands. State-of-the-art models now require significant memory and processing power, often accessible only to large organizations with substantial resources. A growing need, particularly in industry sectors with tight computational and storage constraints, has led to the emergence of “Tiny ML.” This subfield aims to develop models that retain strong performance while fitting into a much smaller memory footprint and computational budget. The gaming industry provides a compelling use case. Many developers have yet to adopt generative LLMs for in-game narrative creation due to concerns about real-time inference on consumer-grade hardware. Tiny ML approaches offer a potential solution by enabling local, on-device, in-engine generation of interactive storytelling content without resorting to expensive cloud-based inference.

This work addresses the problem of achieving high-quality, story-focused text generation from a significantly compressed LLM while maintaining stringent size and compute constraints. Specifically, the goal is to produce a model suitable for interactive narrative generation within games that can fit into approximately 2 GB of memory.

The primary objective of this study is to demonstrate the quality of perplexity, readability, and storytelling on par with larger open-source models widely used, such as the Llama 3.2 3B (11 GB) parameter model, using a compressed version of a Llama 3.2 1B parameter model (4 GB). The approach combines knowledge distillation techniques to preserve or improve its performance in narrative generation tasks and quantization strategies to reduce the size of the model and computational cost. In doing so, this research aims to establish a foundation for practical, resource-efficient generative models that can be readily integrated into computationally limited platforms such as consumer-grade gaming hardware.

II. LITERATURE REVIEW

A. Review of Relevant Literature

In the last calendar year, there has been a consistent push towards small, high-performing LLMs. For example, Microsoft published content about Phi-2 and Phi-1, their tiny LLMs. These models have 2.7B and 1.3B parameters, respectively, and perform well among a variety of standard tests[8]. OpenAI has also exhibited this push, releasing multiple “mini” versions of their popular LLMs for users who need lessened capabilities for particular tasks[12]. These models also perform comparably to their massive counterparts in a variety of established benchmarks. Lastly, there are many engineers currently working on decreasing LLM overhead while keeping a high performance bar. For example, consider[5], a Hugging Face user experiment into their custom tiny LLM that compares a myriad of similarly sized models in multiple benchmarks. Here, we can see that independent groups of engineers can implement methods that could result in evaluation metrics greater than the similarly sized models produced by the biggest players in the space.

While there is substantial work being done in the narrative/game quest generation space [13, 14, 15], we have not found many papers that apply tiny models directly in-game for deployment purposes. We see the intersection of these two problems as a necessity for the future of the generative game space.

III. METHODOLOGY

Our approach is split in two parts: quantization and knowledge distillation. Knowledge distillation is performed on the 1B model with a teacher model represented as the Llama 3B model. Quantization is performed on the Llama 1B and

3B models after distillation. Post-training quantization is the primary technique for quantization.

A. Quantization

The quantization process utilizes post-training quantization via the AWQ library originally inspired by the MIT HAN lab to effectively quantize models with Activation-aware Weight Quantization (AWQ)[7]. The impetus to utilize activation-aware weight quantization is that it is necessary to identify the most important weights within a model. Not all weights contribute equally to the signal a model produces, and it is only necessary to identify 1% (or fewer) of the salient weights within a model. The strongest weights are determined from the activation distribution as opposed to the weights' magnitude. Salient weights are scaled using a per-channel scaling method to minimize quantization error.

Unlike other quantization methods, AWQ does not require backpropagation or weight reconstruction. The technique can adapt across various domains (the research findings demonstrate effective performance for instruction-tuned LMs, multi-modal LMs, and causal LMs). Additionally, only a small calibration dataset is required to demonstrate robust results. The dataset provides representative input samples to the model, allowing it to compute activation distributions for different weight channels. The activation ranges are observed from the calibration dataset and used to calculate the scaling factors and/or zero points. Since no backpropagation is occurring with AWQ, overfitting is avoided and the calibration dataset only collects activation statistics.

In terms of implementation, we set a standard quantization configuration to identify the type of quantization to utilize (e.g., symmetric vs. asymmetric quantization). The AWQ library[3] allows easy implementation and quick calibration time to perform static quantization. Our implementation utilized the standard calibration dataset procured by the research lab to learn the activation ranges. It is possible to leverage other calibration datasets, but the research recommends utilizing the provided calibration dataset.

Our configuration included the standard usage of various arguments tailored towards asymmetric quantization with a standard min/max scale. The weights are scaled appropriately and then pseudo-quantized. Pseudo-quantization simulates the effects of quantization while maintaining the ability to revert to floating-point precision. For example, a value of 5.7 would be converted to an integer (5), then converted back to a float (5.0) to preserve floating-point precision[3]. This is ideal because we can leverage a fully quantized, hardware-friendly model when deployed. However, the model can only run on GPU hardware and loses the ability to run inference on CPU. The entire goal of pseudo-quantization is to gain the performance benefits of quantization without narrowing the deployment options based on a machine's hardware.

B. Knowledge Distillation

Knowledge distillation (KD) is the methodology of training a small(student) model to replicate the behavior of

a larger(teacher) model. By learning from the pre-trained teacher, the student can achieve performance close to the teacher's results while requiring fewer parameters[4]. For our purposes, we employ response-based knowledge distillation, which calculates the teacher-student loss through the cross entropy between the output layer logits of the teacher and student. This form of KD allows the student to not only learn the most likely next token that the teacher would generate but the distribution of all other candidates. Since, in this case, our student model is already pre-trained, we opted for this simple form of KD. Rather than "feature-based" KD which looks back at deeper layers of the network to inform learning.

For each iteration of training, we use a segment of each book and have each model predict the next token in a forward pass. We then evaluate a weighted total loss between the teacher-student loss and the student loss against the true next token.

$$\mathcal{L}_{\text{Total}} = \alpha \mathcal{L}_{\text{KD}} + (1 - \alpha) \mathcal{L}_{\text{CE}}$$

where α is in the range [0-1]. However, we only backpropagate on the student and no gradients are calculated for the teacher.

C. Profiling and evaluation metrics

Note that we are optimizing for performance during inference. In particular, we are optimizing towards a model that is vastly smaller than those currently accessible while maintaining performance to be used in inference. Thus, we are looking at space required to store model and average time per token as measures of model size and latency. The first metric is trivially relevant to the model size and the latter is monitored to verify that the model is fast enough for repeated, consequent use. For performance, we utilize perplexity, readability and number of distinct tokens to evaluate the model. Perplexity is monitored to ensure the model retains a good ability for coherence. Readability is included, as the model outputs are meant for storytelling, where ease of human reading is essential for its perceived proficiency. Lastly, number of distinct tokens helps monitor the ability of the model to produce creativity, which is another commonly assigned feature of storytelling.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

After the models are distilled and quantized, the same evaluation metrics are compared for analysis. Model size, perplexity, readability, distinctness, and average time per token are obtained using identical Torch methods on each model.

B. Performance Comparison of Different Models

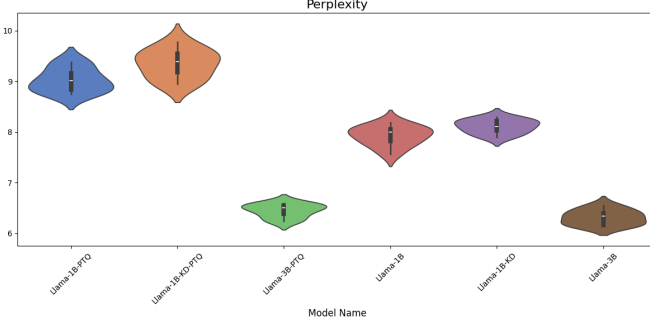


Fig. 1. Violin plot showing perplexity distributions for different models.

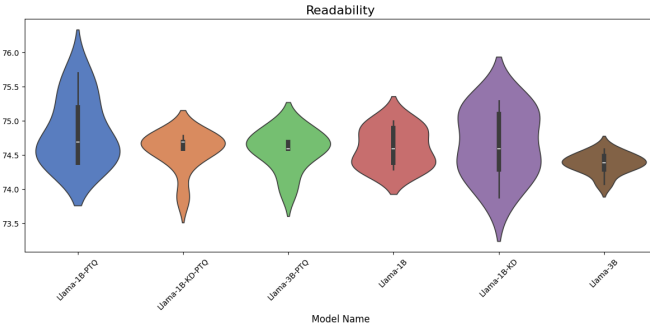


Fig. 2. Violin plot illustrating readability scores across models.

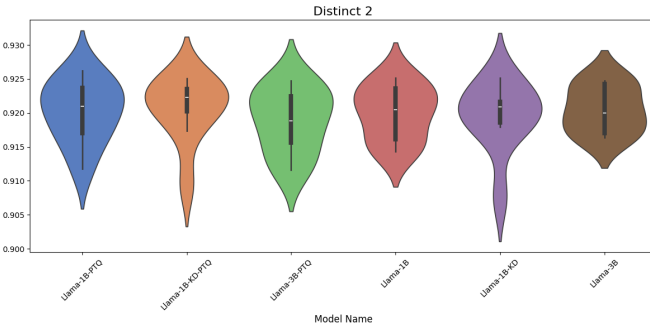


Fig. 3. Violin plot showing token distinctness (Distinct-2) for various models.

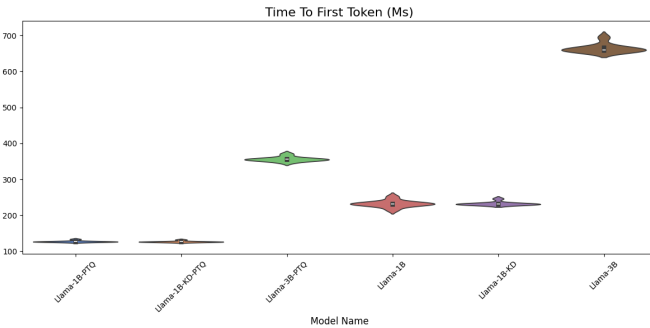


Fig. 4. Violin plot of time to first token (ms) across models.

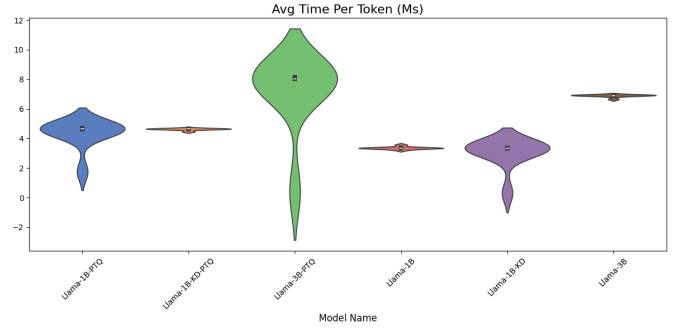


Fig. 5. Violin plot of avg time per token (ms) across models.

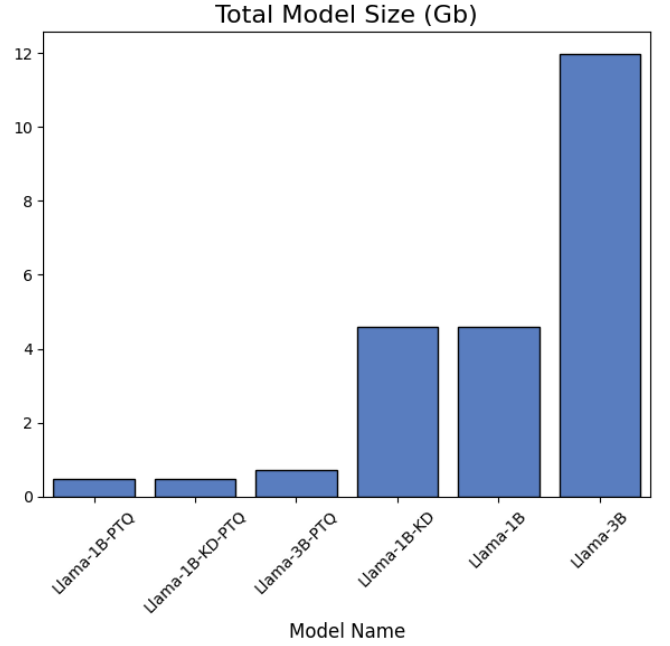


Fig. 6. Plot showing total model size (GB) for various configurations.

C. Analysis of Results

Perplexity is the measure of confidence a language model's predictions. Quantized models, such as Llama-1B-PTQ and Llama-1B-KD-PTQ, exhibit slightly higher perplexity scores (Fig. 1), particularly Llama-1B-KD-PTQ, which has a wider spread of scores. Compression techniques introduce some trade-offs in confidence. For storytelling tasks, this level of confidence is sufficient to maintain narrative quality while benefiting from the significant reductions in size and computational requirements. What we also learn from this plot is that the 3B-PTQ model was resistant to the impact of PTQ over its smaller 1B counterpart.

The readability violin plot in Fig. 2 demonstrates that readability scores are consistent across all models, with even the smallest quantized models performing comparably to the larger ones. Llama-1B-PTQ shows a slightly higher median readability score compared to other quantized models. This

finding is particularly encouraging, as it suggests that size reductions and compression techniques do not negatively impact the clarity of the generated text. For storytelling applications, where readability is a key metric of success, this consistency ensures that even the smallest models can produce text that is clear and engaging to readers.

The distinct-2 metric measures the diversity of tokens in the generated text. Fig. 3 shows the distinct-2 scores across different models. The distribution shows consistent diversity across most models, with slightly higher scores for larger models like Llama-3B. Quantized models, such as Llama-1B-PTQ and Llama-1B-KD-PTQ, exhibit slightly lower diversity, reflecting a modest trade-off due to reduced precision. However, the differences are minimal, indicating that the smaller models still retain reasonable diversity. This result could be an indication of the ability of Tiny LLMs to generate comparably creative and engaging stories with respect to their non-quantized counterparts.

The time-to-first-token (TTFT) plot (Fig. 4) reveals the practical benefits of smaller models, as larger models such as Llama-3B exhibit significantly higher latency compared to smaller models like Llama-1B-PTQ. Quantized models show marked improvements in the TTFT, with the smallest model achieving the fastest response times. This speed improvement is particularly valuable for interactive storytelling applications, where real-time responses are critical for user engagement. While the average time per token (Fig. 5) follows a similar pattern the difference between the large and small models are not as pronounced.

The total model size (Fig. 6) plot further emphasizes the impact of compression techniques. Quantized models such as Llama-1B-PTQ and Llama-1B-KD-PTQ occupy only a fraction of the size of Llama-3B, making them deployable on devices with limited memory. These results highlight the success of the project in achieving its goal of creating a Tiny LLM that balance compactness with acceptable storytelling performance metrics.

V. DISCUSSION

The analysis demonstrates a clear trade-off between model size and performance metrics. While smaller models exhibit slightly higher perplexity and slightly reduced token diversity, their readability remains consistent, and their latency is significantly improved. These findings validate the project’s approach of leveraging quantization and knowledge distillation to develop efficient Tiny LLMs that meet the demands of storytelling applications.

The findings illustrate a stronger case for quantization over knowledge distillation and quantization as effective strategies for bridging the gap between performance and efficiency in Tiny LLMs. Upon further research, we discovered that the combination of these two methods does not always benefit each other. This is due to the regularization effects of KD which diminishes the quantized [representation] [6]. These results provide a solid foundation for future work aimed at optimizing

storytelling-specific tasks and further refining model efficiency for resource-constrained environments.

A. Comparison with Previous Studies

Recent research has demonstrated the effectiveness of model compression techniques, such as pruning, quantization, and knowledge distillation, in creating compact language models that maintain strong performance. Muralidharan et al.[9] explored the combined application of pruning and knowledge distillation to develop compact models with significant size reductions, achieving reductions in memory usage by over 60% while maintaining acceptable accuracy levels on text classification tasks. Similarly, our results show that combining quantization and knowledge distillation can drastically reduce model size to fit within a 2 GB memory footprint, while maintaining coherence and readability critical for storytelling applications. Both studies highlight the trade-offs involved in compression techniques, particularly slight increases in perplexity or minor losses in token diversity. However, these compromises are offset by the practical gains in efficiency and usability on resource-constrained platforms.

Eldan and Li’s work on TinyStories[1] further supports the feasibility of developing compact models tailored for specific tasks, particularly storytelling. Their study demonstrated that models with fewer than 10 million parameters, trained on carefully curated datasets, could generate fluent and consistent narratives for young children. While our study focuses on larger models (1B parameters), the findings of TinyStories align with ours in emphasizing the importance of dataset curation and task-specific optimization in maintaining storytelling quality. Our results extend their approach by applying quantization and knowledge distillation to further compress models while retaining strong performance across perplexity, readability, and token diversity metrics. This allows our TinyLLMs to scale up narrative complexity, making them suitable for broader storytelling use cases, such as interactive gaming.

The findings of this study align with and expand upon previous research by demonstrating that quantization and knowledge distillation are not only effective in reducing model size and memory requirements but also capable of preserving key aspects of storytelling performance. By combining insights from both Muralidharan et al. and Eldan and Li, this work underscores the potential of TinyLLMs to achieve high-quality narrative generation with minimal computational overhead, enabling deployment in resource-constrained environments like consumer-grade gaming hardware.

B. Challenges and Limitations

Some of the challenges that arose during the course of the project include:

- **Different setups on different machines:** Switching between GCP and Terremoto has been an issue.
- **Memory Constraints with CUDA/GPU:** Consistently running out of memory for quantization and knowledge distillation.

- **Storage Constraints:** Consistently running out of storage in VM since each step produces a new model and each model requires ancillary storage.
- **Maintain Accuracy Through Quantization:** For example, basic dynamic quantization did not maintain accuracy very well.
- **Working with HuggingFace:** Getting up to speed with HuggingFace imports and formatting.
- **Properly Computing Metrics:** Code integrations and debugging are complex and take some time to test on each instance.

C. Future Directions

The development of TinyLLMs for storytelling opens up numerous exciting possibilities for future research and applications. This project has the potential to address existing challenges and expand its scope in the following directions:

- **Enhanced Storytelling Capabilities:** Future iterations of this project will focus on improving the model’s storytelling abilities, such as generating more engaging, coherent, and creative narratives. This could involve fine-tuning on datasets specifically curated for storytelling across diverse genres, tones, and cultural contexts.
- **Further Model Size Optimization:** A key goal is to continue reducing the model size while maintaining or even improving its performance in storytelling tasks. For example, structured pruning could identify and remove less important attention heads, which would reduce the overall computation required while maintaining text coherence.
- **Developing a Storytelling Benchmark:** To better evaluate storytelling performance, a dedicated testing dataset or prompt suite could be developed. This benchmark would measure key metrics such as narrative coherence, readability, diversity, and emotional engagement, providing a standardized framework for comparison with other models.
- **Training on More Diverse and Specialized Datasets:** Future efforts could incorporate datasets with a wider variety of storytelling styles, cultural perspectives, and linguistic diversity. Specialized datasets, such as children’s stories, interactive dialogue scripts, or fantasy world-building content, could tailor the model for specific storytelling needs.
- **Practical Applications and Deployment:** Deploying the TinyLLM in resource-constrained environments, such as mobile devices, game engines, or embedded systems, would demonstrate its utility in real-world applications. Specific use cases include interactive storytelling in video games, personalized bedtime stories for children, or AI-assisted content creation for independent authors.
- **User-Centric Customization:** Enabling end-users to customize the storytelling style, tone, or content themes through lightweight fine-tuning or prompt engineering would make the TinyLLM more versatile and appealing to a broader audience.

VI. CONCLUSION

A. Summary of Findings

This study demonstrates the feasibility of compressing LLMs for high-quality storytelling while addressing computational and memory constraints. By employing quantization and knowledge distillation techniques, we reduced the Llama 3.2 1B model to under 2 GB, achieving significant improvements in latency and deployability without compromising critical storytelling metrics.

Overall, the findings validate the proposed methods for creating resource-efficient TinyLLMs. While both quantization and knowledge distillation contribute to achieving a balance between size reduction and storytelling quality, the results highlight that quantization alone offers a stronger and more consistent impact on bridging the gap between performance and efficiency. Further analysis revealed that the combination of quantization and knowledge distillation does not result in significant improvements, as the regularization effects of knowledge distillation are diminished when converted to a quantized representation [6]. This limitation indicates room for improvement in our method, particularly in exploring a quantization-aware knowledge distillation process to better align the two strategies.

B. Recommendations for Future Research

Future research should focus on exploring other compression techniques beyond post-training quantization and knowledge distillation, such as structured pruning or hybrid quantization methods, to achieve even greater efficiency while maintaining storytelling quality. Additionally, the development of task-specific benchmarks for storytelling, including metrics like narrative coherence, emotional engagement, and creativity, would provide standardized evaluation frameworks for further model improvements. Cross-linguistic and cultural adaptation is another promising direction, enabling Tiny LLMs to cater to diverse languages and storytelling traditions by incorporating broader datasets. Finally, efforts should be made to optimize real-time deployment on edge devices, addressing latency, energy efficiency, and compatibility with consumer-grade hardware, which would expand the practical applicability of Tiny LLMs in interactive storytelling and gaming environments.

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